# An Open Source Rendering Pipeline for Neuroimaging Data

## INTRODUCTION

Despite the existence of high resolution technologies for 3D visualisation, imaging data seems confined to proprietary software linked to DICOM and NIFTI files shipped by MRI equipment manufacturers. An issue that arises from this is that many of these pieces of software fail to utilise modern rendering solutions that can drastically increase the quality of final renders for use in scientific communication and publication, or are wholly confined to 2D stacked image representations. This project aims to describe a methodology that retains the stability and flexibility of the proprietary programs, but uses open source intermediates and modern 3D software to convert MRI information into truly 3D topology that can be rendered in a path traced engine. The aim is to allow for more efficient and effective scientific communication and outreach of neuroimaging data, and for entities without access to costly proprietary software to create high-quality figures for scientific communication and education.

## METHODS

### *Data Acquisition and Formatting*

By its very nature MRI data is often protected information and can be difficult to obtain. Within this report the Yale Longitudinal Dataset of Brain Metastases on MRI with Associated Clinical Data *(Chadha et al, 2025)* was used for versions 1.0 and 1.1 due to their generous open access policies. For version 2.0 onwards, the “Olfactory dysfunction and functional connectivity changes in cognitively normal Parkinson’s disease” published on OpenMRI is used (*Yoneyama et al, 2018*) due to its significantly higher resolution (<20 versus 255 z-levels respectively). This workflow will function for any clinical data obtained in the DICOM or NIFTI file formats.

The example render used in v1.0 and v1.1 is Patient ID YG_04YGLO8ATWRL, from 2013-07-25 using the PRE scan information. This dataset, alongside many other imaging sets, provides it’s data in the NIFTI format to preserve patient anonymity and trim out excess information usually preserved in a DICOM file. Unfortunately, InVesalius (the program we use to create the 3D object necessary for the final render) only accepts imports in a DICOM format. Therefore, we must convert the NIFTI file back to DICOM. This is done via the python program nii2dcm *(Roberts, 2025)*. For further instruction, consult the documentation found on the nii2dcm repository.

### *DICOM Masking and .STL Conversion*

Now in the DICOM file format, we import the folder of stacked images into InVesalius *(Amorim et al, 2015)*. Following a quality check, the images are masked in order to select desired tissues and remove noise. For the purposes of the example render, the default range preset was used. After masking, the generated 3D information can be exported as an .STL file for import into Blender *(Blender Team, 2025)*.

Alternatively to automatic masking, I would strongly recommend using the manual masking feature in InVesalius. Changes seen in v1.1 and v2.0 reflect the increased detail seen from taking the time to mask tissue data manually in order to preserve ventricular detail.

### *Cleaning and Rendering in Blender*

Here the steps taken to create the final animated render are described in brief, however as many bioscientists have limited familiarity with blender a template file is provided in the project repository. Upon loading the template file you must import the generated .STL of the MRI data and align it with the rectangle labelled “Occlusion Plane” by placing it approximately above the origin.

After this, I recommend cleaning your mesh both manually and using the tools available in Blender. Manually remove any artefact mesh data present outside of the brain model. Next I would advise decimating the mesh to an appropriate scale factor (e.g. 0.5) to allow dynamic recalculation of topology as the occlusion plane moves through the object. Note that this will modify the raw MRI information, technically rendering it less accurate. It will, however, significantly reduce render times and improve visual fidelity. Finally, create a Boolean intersect between the occlusion plane and imported anatomical data, but do not apply the modifier. Crucially, this modifier must be placed onto your imported mesh and have the “allow holes” and “complex calculator” options checked. Here, you can render your final animation or images and optionally apply any materials you wish to your mesh to alter its visual properties.

You may also wish to create your own template file, or modify the one included in the repository. In this case after completing topographic cleanup a camera was created and aligned with the desired final angle. Bidirectional lights were also added. Keyframes were added to the cube, and it was marked non-visible in the render in order for it to occlude the brain mesh without showing.

### *fMRI BOLD Visualisation*

To co-visualise fMRI BOLD data alongside anatomical data the desired frame of the BOLD data must be imported into InVesalius and converted into a mesh using the DICOM masking method detailed above. During this process, you must decide on a masking threshold to determine what level of activity should be visualised. This will be study and machine specific. For the example in this report any value > 950 was included in the mesh.

After this import the STL object into Blender and align it with the anatomical data. It is recommended to use a semi-transparent material for the anatomical data to better view the BOLD mesh.

## DISCUSSION

With v2.0, the pipeline has demonstrated itself capable of producing much higher quality 3D renders whilst maintaining relatively low render times (mean 20s per frame at 2160x2160 with an 82 FP32 TFLOPS and 191 FP16 TFLOPS system). As such, I believe it is now in a suitable place to begin iterating and expanding out to other areas of imaging such as CT and OCT data.

Whilst the initial expansion of the pipeline into visualising fMRI BOLD data appears promising, it faces two significant challenges. The first is that it is incapable of rendering a true heatmap of BOLD data, instead only rendering a mesh above a threshold. This could theoretically be solved by either SPM integration or a custom scripting or geometry nodes solution, but both would be very time consuming to implement and so will be delayed until further improvements have been made. The second issue is that currently only a single frame of the 198 included BOLD frames for this patient are rendered here. I believe this may be resolvable through new import node system implemented in Blender 4.5, and will attempt to implement this in the next iteration of the project.

Currently under investigation as a significant potential improvement would be integration of SPM to allow for more accurate topological exports. This program would also fit the aim of this pipeline as it is open source. For a list of further improvements and expansions to the project I aim to make in the more distant future, please consult the repository README.

In the meantime, I hope that this provides a method of creating visually appealing renders of anatomical MRI information for scientific communication through a methodology which is entirely free and open source. My sincere acknowledgements to the creators of Blender, InVesalius and nii2dcm for creating the invaluable tools used in this pipeline, and the creators of OpenMRI for dataset access.

## REFERENCES


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